SWy-3 montmorillonite (1 %, 22 °C) — provisional polarized DLS analysis report

Executive Summary

Dataset. SWy-3 Wyoming montmorillonite clay in water at 22 °C by multi-angle PCS: 14 angles (20–150°), three measurements per angle — depolarized VH, polarized VV (both 2D pseudo-cross), and 3D modulated cross, 10 repetitions each. The full suite runs on VV.

Main result. A flocculated Wyoming-montmorillonite clay suspension — anisotropic platelets aggregated into broad, mass-fractal flocs in water.

- **Size:** the scattering is dominated by ˜**507 nm** flocs (intensity-weighted); by number the population centres near ˜**56 nm**; z-average R_h **203 nm**; broad / polydisperse (mean PDI 0.52).
- **Structure:** the static Porod slope I $\propto q^{-2.44}\pm0.02 \Rightarrow$ **mass fractal / branched (D≈2.4)** — a restructured / branched floc, not compact spheres, so the sphere PSD form factor is only an effective approximation and $\rho = R_g/R_h$ **is not determined**.
- **Anisotropy:** a depolarized signal $I_{\mathrm{VH}}/I_{\mathrm{VV}} \approx 1.3\%$ confirms plate-like, optically anisotropic scatterers; rotational diffusion is **not credibly resolved** (carried only by the weak VH channel).
- **Measurement limits:** the lowest q still has $q{\cdot}R_g \approx 1.8$ (> 1.3), so no Guinier regime is reached — R_g, aggregation number and ρ are not directly measurable; the toluene absolute-intensity scale is unverified.

Confidence. The z-average r_h carries replicate/angle bootstrap intervals; leave-one-angle-out cv gives predictive r^2=0.860; the corenn inversion converged.

Assumption-sensitive. Viscosity, refractive index, q-from-angle calibration, the toluene/absolute-intensity reference, the assumed sphere form factor, and the CORENN regularization all enter the reported sizes; see the Assumption Register.

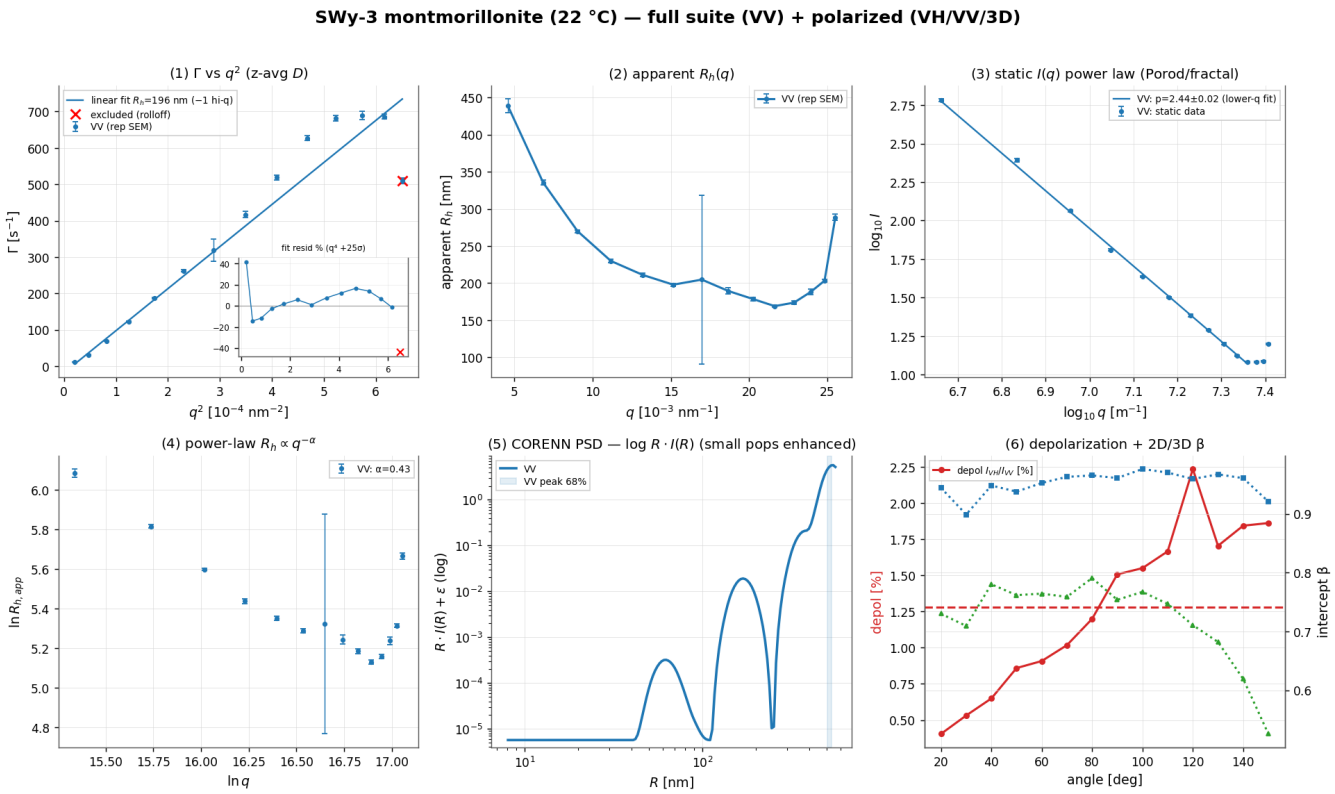
Recommended next action. fit a mass-fractal (and/or oblate-disk) form factor — the Porod slope rules out compact spheres; confirm the toluene reference to lock the absolute SLS scale.

> **Provisional flags:** toluene/absolute-intensity reference not confirmable from the export; lower-q Porod/fractal slope $q^{-2.44}\pm0.02$ implies a mass-fractal / branched floc, not the sphere assumed by the PSD; high-q SWy3 points flatten/upturn and are excluded from the slope as an end-of-scaling/background diagnostic; the clay is optically anisotropic (plate-shaped) — the sphere/translational model is an approximation.

Summary Table

quantity	value
angles × channels	14 × (VH, VV, 3D)
z-average R_h (all-q slope) [68%/95% CI]	203 [68% 180–227; 95% 167–255] nm
apparent D (linear Γ vs q^2, rolloff excl.)	D=1.16e-12±4.3e-15 m^2/s → R_h=196 nm (linear fit, 13/14 angles, -1 hi-q rolloff at q≤0.0248 nm^{-1})
shape $\rho = R_g/R_h$	not determined (Porod regime)
apparent R_h low→high q	387→246 nm
static power-law I∝q^{-p}	p=2.44±0.02 (r^2=0.999, q=0.00459–0.0229 nm^{-1})
implied morphology (Porod)	mass fractal / branched (D≈2.4)
platelet radius a = (π/2)·R_h (oblate-disk SE)	primary ≈ 88 nm; floc-eff ≈ 308 nm
Γ(q^2) predictive R^2 (LOO-CV)	0.860 (in-sample 0.911)
CORENN eff.DOF / cond	18/24, κ=6e+13
CORENN multi-angle PSD	507 nm (96%), 350 nm (4%)
CORENN peak [68%/95% CI, bootstrap]	511 [68% 505–535; 95% 498–538] nm
6 D_r (VH-VV intercept)	154.8 /s (not credibly resolved)
depolarization ratio	1.28 %
mean intercept β (2D / 3D)	0.95 / 0.72

Consolidating Figures



Consolidating figure: 5 suite panels (VV) plus the depolarization ratio and 2D/3D coherence β vs angle (raw-derived). Error bars are 1 SEM across repetitions where repetitions exist; the static exponent is fit only on the accepted lower-q scaling window.

Primary Model Results

Channel identification. Per angle, the two pseudo-cross measurements are VH (low count rate) and VV (high count rate); the third is a 3D modulated cross. q is recomputed from the angle. The full static+dynamic suite runs on the polarized VV channel.

Step 1 — per-angle cumulants → Γ(q) (VV). A 2nd-order cumulant on the initial decay (g1_min=0.6) gives the decay rate at each angle; 14 angles are kept.

Step 2 — Γ vs q² (apparent diffusion). Γ vs q^2 is linear over the bulk but the **1 highest-q angle(s) roll off** (Γ plateaus/drops at q = 0.0255 nm^{-1} — weak signal, number fluctuations, and residual multiple scattering at high angle), so they are excluded. A straight-line fit $\Gamma = c + D \cdot q^2$ to the remaining **13 of 14 angles** ($q \leq 0.0248$ nm^{-1}, σ from replicate SEM) gives the **apparent diffusion coefficient D = 1.156e-12 ± 4.3e-15 m^2/s → R_h = 196 nm** (vs 203 nm for the all-angle fit that includes the rolloff). The fitted intercept c ≈ -18/s is non-zero — a low-q curvature / non-diffusive offset (large-aggregate slow mode, baseline, or weak S(q)) — so read D as the slope of the linear region, not Γ/q^2. **The fit residuals are structured, not random** (figure inset): they trend systematically with q (Spearman resid–q = +0.49; runs 4 vs ˜7 expected) and a q^4 curvature term is significant at **+25**σ — so even with the rolloff removed, Γ vs q^2 genuinely **curves upward** — D_{app} rises with q, the signature of **polydispersity**: the line is a chord and no single slope is *the* diffusion coefficient. The q-dependent apparent size (next step) and the CORENN PSD are the proper description.

Step 3 — apparent R_h(q) (polydispersity). apparent R_h falls with q, from **387 nm** (low q) to **246 nm** (high q) (curvature C=7.75e-16, Spearman -0.58); mean PDI 0.52 — a broad / multi-modal population (large-particle tail at low q).

Step 4 — static intensity I(q), regime from the PSD. The CORENN PSD gives dominant scatterers of R ≈ 500 nm ($R_g \approx 387$ nm, sphere-derived from the PSD radius — no Guinier regime is reached, so this is a model estimate, not a measured R_g), so q·R_g spans 1.8–9.9 across the angles → **form-factor / Porod (q·R $\gtrsim \pi$ throughout)**. On the 11 point contiguous lower-q scaling window (q=0.00459–0.0229 nm^{-1}): $I \propto q^{-2.44} \pm 0.02$ → **a mass fractal — a branched / randomly-branched / aggregated network, mass-fractal dimension D≈2.4** (r^2=0.999). The 3 higher-q point(s) are excluded from the exponent because they show flattening followed by an upturn relative to the lower-q scaling law; physically this is the end of the mass-fractal / form-factor scaling range plus the form-factor rise of a smaller, weakly-scattering population — an *upturn* rather than a plateau is the signature of small particles, consistent with the small-R modes that appear under number-weighting in the size-distribution section below, not a second Porod law.

Step 5 — dynamic q-dependence. The apparent size follows a power law $R_{h,app} \propto q\hat{}\text{-}\alpha$ with α=0.43 (r^2=0.65) — a compact descriptor of the breadth of the distribution ($\alpha \to 0$ monodisperse, $\alpha \to 1$ fully scale-free).

Step 6 — CORENN multi-angle PSD (forward model). The joint inversion of the resolvable angles, on a radius grid bounded by dynamic observability plus the static q-window (PSD grid 8–560 nm, upper bound set by dynamic) and the main-relaxation lag window (λ=1.4e+00), recovers one q-consistent intensity size distribution: peak(s) at **507 nm (96%), 350 nm (4%)** — this peak sits at the PSD upper observability bound (˜560 nm); the true material may extend beyond the measured lag/q window, so treat it as an unresolved slow-tail / lower-bound signature rather than a resolved peak.. An 20-angle bootstrap of the inversion puts the primary peak at 511 nm [68% 505–535; 95% 498–538 nm].

Static Regime And Shape

The CORENN PSD anticipates which static regime each angle probes (Guinier where q·R_g < 1.3, Porod where q·R $\gtrsim \pi$). When a **valid Porod slope** exists, its exponent informs the form-factor / morphology model (q^{-4} a smooth sphere; 3<p<4 a surface fractal, D_s=6-p; p≈3 a rough-surface or branched network; 2≤p<3 a mass-fractal / branched aggregate, D=p; p≈2 a coil; p≈1 a rod). $\rho = R_g/R_h$ is the model-independent shape check (sphere ≈ 0.78, coil ≈ 1.5, rod $\gtrsim$ 2), determined only when a static Guinier R_g exists.

condition	q·R_g span	regime	static fit	implied morphology	$\rho = R_g/R_h$
22 °C	1.8–9.9	form-factor / Porod (q·R $\gtrsim \pi$ throughout)	Porod/fractal I$\propto q^{-2.44}\pm$0.02, q=0.00459–0.0229 nm^{-1}	mass fractal / branched (D≈2.4)	not determined

Particle model. The PSD inversion assumes a homogeneous-sphere form factor (which predicts a Porod slope q^{-4}). The measured static power law instead implies — 22 °C: a mass fractal — a branched / randomly-branched / aggregated network, mass-fractal dimension D≈2.4. The sphere form factor is therefore **inconsistent with the static data**: read the PSD peak as the *effective size* of a rough / branched aggregate rather than a compact-sphere radius, and prefer a fractal (surface- or mass-fractal) form factor for a quantitative size. $\rho = R_g/R_h$ is not independently determined here, so it cannot further constrain the shape.

Error Propagation And Sensitivity

Headline R_h and the PSD peak carry bootstrap intervals (ρ where determined). The **replicate** bootstrap is measurement repeatability; the **angle/model** bootstrap is robustness to angle selection — the dominant term for polydisperse samples.

condition	R_h replicate 68%	R_h angle/model 68%	PSD peak 68%	ρ 68%
22 °C	201–204 nm	180–227 nm	505–535 nm	—

Systematic sensitivity (not included in the intervals above). $R_h = \mathrm{kT}/(6\pi\eta\mathrm{D})$ scales inversely with viscosity, so a viscosity error propagates linearly into R_h; the q calibration (scattering angle + refractive index) shifts every q-dependent quantity; the absolute-intensity (toluene) scale sets the SLS magnitude; and the assumed sphere form factor biases the PSD radii for non-spherical particles. See the Assumption Register.

Fit Quality And Model Adequacy

Beyond r^2: a hold-out cross-validation (predictive R^2), a residual-structure test (runs / lag-1 autocorrelation — structured residuals flag model inadequacy even at high r^2), and the CORENN inversion's convergence and conditioning.

condition	Γ-vs-q^2 R^2 (in / CV)	Γ resid	CORENN g2 resid	CORENN: conv / GCV / κ / eff-DOF
22 °C	0.911 / 0.860	**structured**	**structured**	yes / interior / κ=6e+13 / 18/24

Structured Γ-vs-q^2 residuals reflect the known polydispersity curvature (the single slope is a summary, not the model); structured CORENN g2 residuals flag form-factor / single-distribution imperfection. A predictive CV R^2 well below the in-sample R^2 indicates over-fitting / poor extrapolation across angles.

Conditioning: $\kappa \approx$ 6e+13 means the inversion is severely ill-conditioned (˜14 of ˜16 double-precision digits lost), so the recovered distribution is fixed by the regularization, not the data alone — reinforcing the modality caveat. The eff-DOF/n_{basis} above is the multi-angle inversion's data-supportable complexity; it is a different inversion from the single-angle stitched one whose eff-DOF (˜5–6) appears in the size_max diagnostic.

Sample Synthesis

What the independent observables imply about the sample as a colloid — derived only from the scattering data above, with the inversion caveats carried through:

Physical picture. The scattering is dominated by ˜**539 nm** objects (intensity-weighted); by number the population centres near ˜**56 nm**, and the z-average R_h is 203 nm — best read as one broad, polydisperse distribution

spanning these scales rather than well-separated species (see the size-distribution-by-weighting and size_max sections).

Hydrodynamic shape (Guinier–Porod → Stokes–Einstein). The static slope identifies the dominant scatterer as **a mass-fractal aggregate (D≈2.4)**, so the geometry-matched Stokes–Einstein relation is $\mathbf{D = kT/(6\pi\eta R_h)}$ **(Stokes radius)** — the effective Stokes radius R_h is **196 nm**. That ˜2D slope is geometrically degenerate, so **a thin oblate disk / platelet** is equally consistent — then $D = kT/(12\eta a) \Rightarrow a = (\pi/2)R_h \rightarrow$ the disk radius a is **308 nm**. A branched aggregate has no closed-form Stokes–Einstein, so R_h is an effective Stokes radius; the same ˜2D slope is also consistent with a flat platelet building block (disk SE).

Floc structure & aggregation mechanism. The static Porod slope gives a mass-fractal dimension $D_f \approx$ **2.4**, which points to **a restructured / compacted aggregate — denser than simple RLCA (internal reorganization, or for plates an edge–face 'house-of-cards' association)**. Taking the ˜56 nm number-mode as the primary unit, the fractal scaling $N \approx (R_{floc}/R_{primary})\hat{}D_f$ gives an order-of-magnitude **aggregation number N ˜ 250** primary units per ˜539 nm floc (geometric, and sensitive to the primary-size choice and the inversion).

Primary unit. The number-weighted mode (˜56 nm) is the smallest resolved scatterer — plausibly the primary particle or smallest aggregate. A depolarized signal ($I_{VH}/I_{VV} \approx 1.3\%$) shows the scatterers are optically anisotropic (plate- or rod-like primaries), but rotational diffusion is not credibly resolved (the depolarized fit gives D_t(VH)=6.2e-13 m^2/s, 0.6× the polarized D_t(VV)=1.1e-12 — inconsistent for one translational mode; the depolarized channel is weak ($I_{VH}/I_{VV} \approx 1.3\%$)), so an independent shape/size from rotation is not available here.

Not determinable from this angular range. The lowest q still has $q \cdot R_g \approx 1.8$ (> 1.3), so the **Guinier regime is never reached**: an independent radius of gyration, an absolute molar mass / aggregation number, and $\rho = R_g/R_h$ cannot be measured here — only inferred. Reaching them needs $q \lesssim 0.00336$ nm^{-1} (smaller scattering angles, low-q SLS, or SAXS).

Caveats that bound these conclusions. *(1) Dilute limit / S(q):* the analysis assumes single-particle scattering; if the suspension is not dilute, a low-q structure factor (interparticle correlations or incipient gelation) can bias the Porod/fractal slope and the apparent sizes — a dilution series would test it. *(2) Inversion dependence:* the modes and the aggregation number inherit the ill-posedness and size_max / λ / form-factor sensitivity documented in the size-distribution sections. *(3) Stability:* this is a single time-point of a system that can aggregate or sediment; a repeat / time-resolved run (or checking the count rate for drift across the angle scan) is needed to establish the distribution is stationary.

Consistency across observables. The broad dynamic distribution (mean PDI 0.52, apparent R_h falling with q) matches the polydisperse fractal floc inferred from the static slope; the non-sphere Porod slope and the undetermined ρ agree (both say 'not a compact sphere'). No single observable here is contradicted by another — the picture is one broad, anisotropic, polydisperse population rather than a set of distinct species.

Mode-Specific Diagnostics

Rotational diffusion (depolarized VV/VH)

Polarized and depolarized translational diffusion: D_t(VV)=1.12e-12, D_t(VH)=6.15e-13 m^2/s. The VH-VV intercept difference $6D_r$ = 154.8 /s makes rotation **not credibly resolved**: the depolarized fit gives D_t(VH)=6.2e-13 m^2/s, 0.6× the polarized D_t(VV)=1.1e-12 — inconsistent for one translational mode; the depolarized channel is weak ($I_{VH}/I_{VV} \approx 1.3\%$). These are sub-$\mu$m clay flocs, for which whole-floc rotation ($D_r \propto 1/R^3$) is intrinsically too slow to resolve; the individual platelets are strongly anisotropic but their rotation is carried only by the weak depolarized channel.

Optical anisotropy (depolarization ratio)

The depolarized ratio I_{VH}/I_{VV} averages **1.28 %** (0.40–2.24 % across angles). Montmorillonite platelets are optically anisotropic (form birefringence of a thin disk), so a non-zero VH signal is expected; the ratio quantifies the anisotropy of the scattering objects (single platelets vs. more isotropic flocs) and is the robust polarization observable here.

Platelet hydrodynamics (oblate-disk Stokes–Einstein)

The Guinier–Porod slope is dimensionally ˜2D and the depolarized channel shows optical anisotropy — both pointing to **thin plates** (consistent with montmorillonite's platelet habit), so the sphere relation $R_h = k_B T/(6\pi\eta D)$ is not the natural size. For a thin **oblate disk** of radius a the orientation-averaged translational diffusion is $D_t = k_B T/(\mathbf{12}\eta\mathbf{a})$ — a disk diffuses like a sphere of radius $R_h = (2/\pi)a$, so the **platelet radius a = $(\pi/2)\cdot R_h \approx$ 1.57** R_h (tumbling rotation $D_r = 3k_B T/(32\eta a^3)$).

- **Individual platelets** (number-weighted primary, $R_h \approx 56$ nm) → disk radius **a ≈ 88 nm** (˜177 nm lateral) — in the montmorillonite platelet-size range, and the morphology for which the thin-disk law is rigorous.
- **Dominant scatterers** (z-average $R_h \approx 196$ nm) → effective disk radius ≈ **308 nm**, but these are **fractal flocs ($D_f \approx$ 2.4), not single platelets**, so neither sphere nor thin-disk is exact; the sphere R_h stays the conventional Stokes radius and the disk value is only an effective size.
- **Rotation (depolarized).** Taken at face value the measured $6D_r = 154.8$/s ($D_r \approx 25.8$/s) gives a disk radius ≈ **249 nm** via $D_r = 3k_B T/(32\eta a^3)$; but rotation is **not credibly resolved** (weak VH channel), so this is illustrative, not a measurement.

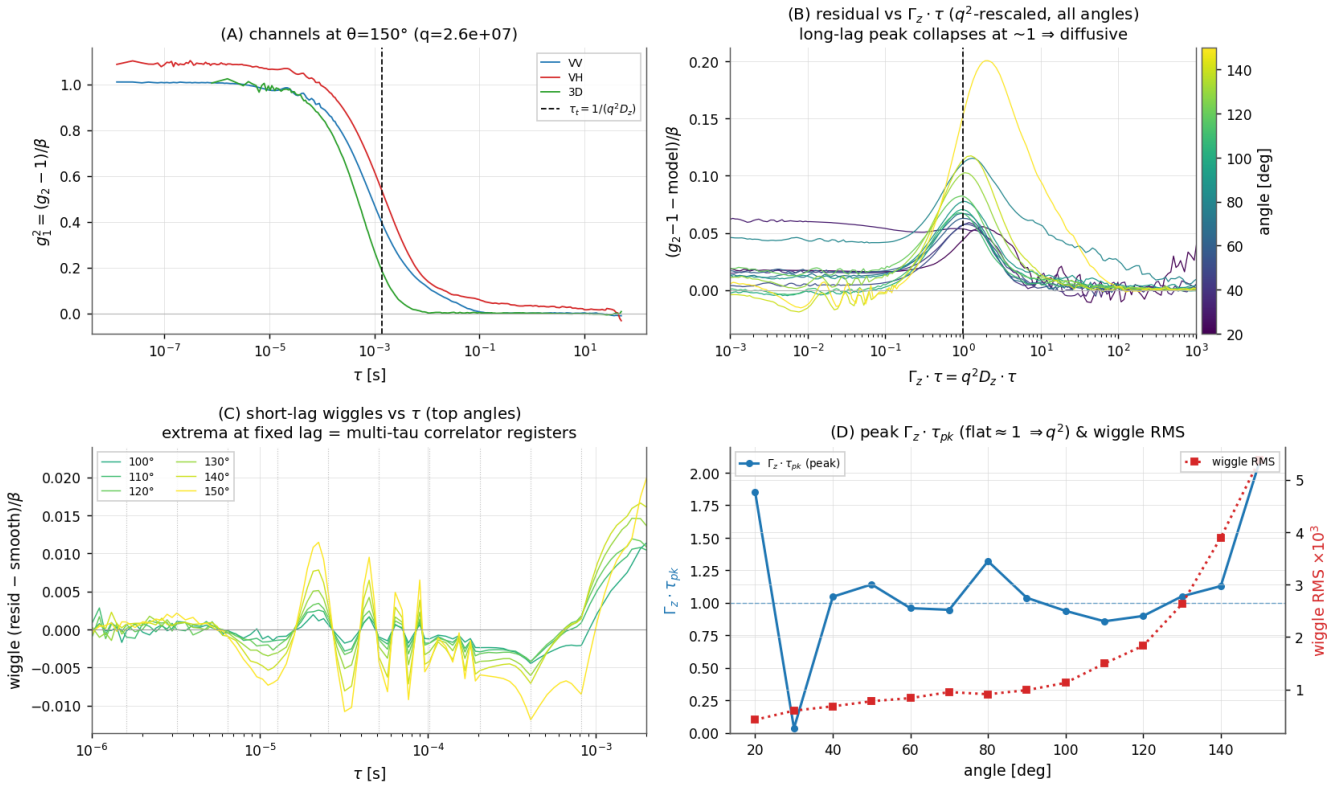
The thin-disk Stokes–Einstein is the right morphology for the **primary platelets** (˜88 nm radius); the micron-scale flocs that dominate the scattering are fractal aggregates best described by the sphere R_h.

2D vs 3D cross-correlation (multiple scattering)

The 3D modulated cross-correlation suppresses multiple scattering, important for a turbid clay suspension. The coherence factor (intercept β) averages 0.95 for 2D VV vs 0.72 for 3D; their angular dependence (consolidating panel 6) flags where multiple-scattering / geometry corrections matter.

High-angle correlogram diagnostics (μs–ms ripples)

The high-angle single-exponential residual contains **two distinct features** that separate by their q-scaling: a **long-lag peak** that shifts to shorter lag with increasing angle, and **short-lag wiggles** that barely move. Panels (B)–(D) tell them apart across all angles.



Reading the panels. **(A)** VV/VH/3D at the top angle (a faster VH would be the rotational signature $\Gamma_VH = q^2 D_t + 6D_r$; a VV-vs-3D split flags modulation / multiple scattering). **(B)** residual vs the q^2-rescaled lag $\Gamma_z{\cdot}\tau$ — a feature that **collapses onto $\Gamma_z{\cdot}\tau \approx 1$** for every angle is a genuine diffusive relaxation. **(C)** the short-lag wiggle component vs *absolute* τ — extrema that line up at **fixed lag** (dotted: the multi-tau correlator register boundaries) are instrumental, not q-dependent dynamics. **(D)** trends: a flat $\Gamma_z{\cdot}\tau_pk \approx 1$ confirms the peak scales as q^2; the wiggle RMS grows as the signal weakens at high q.

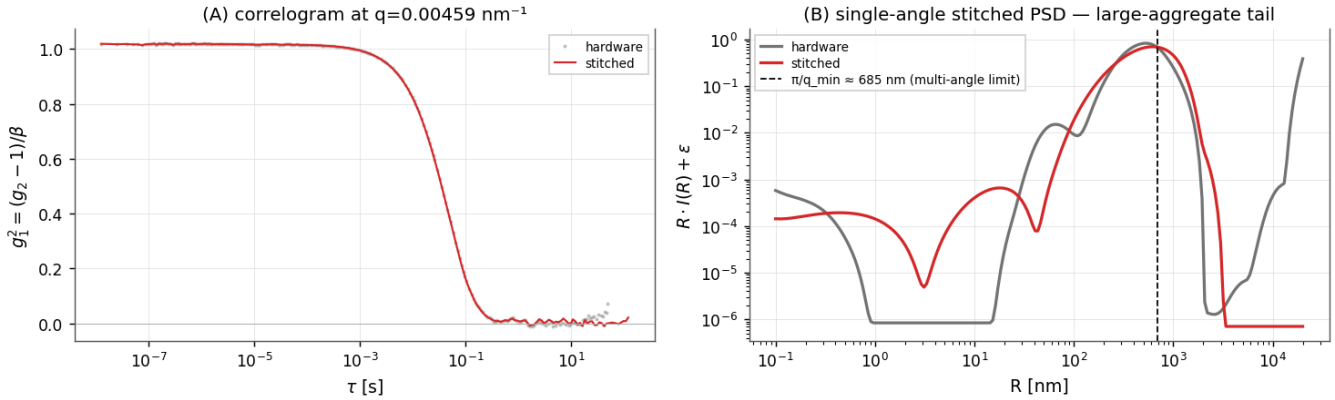
This sample — the two features separated across all 14 angles: **(1) the long-lag peak** sits at $\Gamma_z{\cdot}\tau_pk \approx 1.04$ and stays roughly constant across angles (panel D), i.e. it tracks $\tau_pk = 1/(q^2 D_z) \propto 1/q^2$ and collapses onto a single curve under q^2 rescaling (panel B) — a **genuine diffusive multi-mode / polydispersity** deviation (the broad relaxation-time distribution of the sample), strongest at high q. **(2) the short-lag wiggles** pin to *fixed* absolute lags (≈ 13, 26, 51, 102 μs), independent of angle and locked to the multi-tau correlator's register boundaries (panel C), with the amplitude growing ≈13× from low to high angle as the signal weakens — a **correlator / instrumental artifact**, not sample dynamics.

Channel check (A): at the top angle $\Gamma_VH/\Gamma_VV = 0.68$, i.e. VH decays slower — not a rotational signature; but the depolarized channel is weak ($I_{VH}/I_{VV} \approx 1.3$ %) so Γ_VH is noise-limited.

Slow-mode / large-aggregate tail (single-angle stitched)

The multi-angle intensity PSD is q-resolution-limited: its R^6-weighted q→0 representation, coupled through the sphere form factor, edge-piles for $R \gtrsim \pi/q_{\min} \approx$ **685 nm**. A *single-angle* CORENN inversion of the gap-aware **stitched** correlogram (clean long-lag pooled over the full run, τ_max 50→123 s) instead recovers the slow diffusive modes directly from the decay-rate spectrum — with a data-bounded amplitude, so it resolves the large-aggregate tail where the multi-angle PSD cannot.

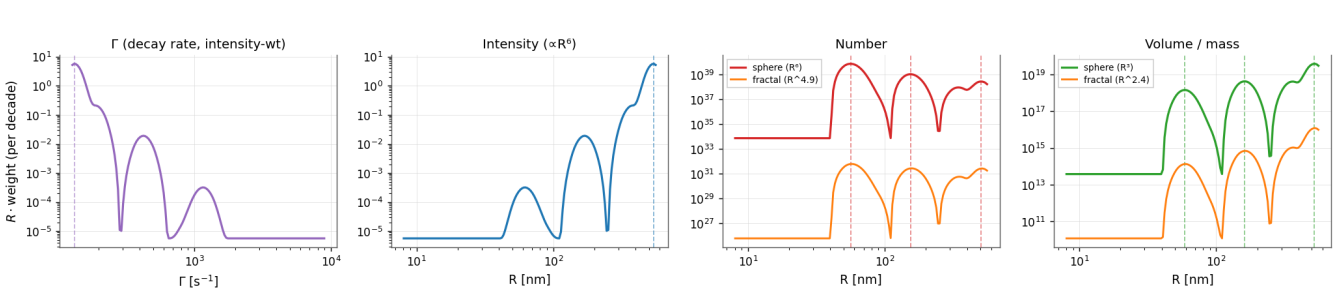
Slow-mode / large-aggregate tail (single-angle stitched) — SWy-3



At the lowest angle (q=0.00459 nm^{-1}), where large aggregates scatter most strongly (forward, $\propto R^6$), the stitched single-angle inversion finds **34% of the intensity beyond** $\pi/q_{\min}$ (vs 27% from the noise-dominated hardware long-lag), extending to ˜1190 nm — confirming and sizing the large-aggregate tail the multi-angle PSD must omit. Read it as a *hydrodynamic* (slow-diffusion), intensity-weighted view: these aggregates dominate the low-angle intensity even though they are a small *number* fraction, so this is the resolvable home of the aggregate mass — not a contradiction of the few-hundred-nm q-consistent main population in the multi-angle PSD.

Size distribution by weighting — continuous summary, then apparent modes

DLS recovers a size distribution by an **ill-posed Laplace inversion**: many different distributions — a smooth continuum or a set of discrete modes — reproduce the measured correlation function equally well within noise, and the regularized inversion (CORENN) returns just one of them, fixed by its smoothness penalty, radius grid, and form-factor assumption. The number and sharpness of peaks are therefore properties of the inversion, not facts the data establish on their own. The report gives **two views**: (1) assumption-light continuous percentiles, and (2) the inversion's apparent modes/shoulders, flagged as inversion-dependent and trustworthy only where they survive the size_max / λ sensitivity tests. CORENN returns the *same* inversion under every weighting (the vendor's Γ / Intensity / Number / Volume tabs); re-weighting only relocates and re-weights the features.



Γ (decay-rate) view — the rawest signal (figure panel 1). Before any size model, the dominant intensity component relaxes at $\Gamma \approx$ **134 s^{-1}** (relaxation time $\tau = 1/\Gamma \approx$ **7.49 ms**) at the reference angle; the size columns below are this spectrum mapped through Stokes–Einstein $\Gamma = k_B T\ q^2/(6\pi\eta R)$.

Continuous-distribution summary (percentiles; no modes assumed). The assumption-light description — percentiles of the recovered distribution, robust to how it is partitioned into modes. Intensity is narrow (dominated by the largest scatterers); number-weighting ($\div R^6$) strongly amplifies the inverted PSD's small-R end, so its percentiles are the least certain.

Weighting	D10 (nm)	D50 (nm)	D90 (nm)	D_g (nm)	σ_g (GSD)	span	log-skew
Intensity ($\propto R^6$)	442	510	550	496	1.14	0.21	-4.39
Number (fractal, $R^{4.9}$)	52	68	486	113	2.31	6.36	+0.72
Number (sphere, R^6)	50	58	152	69	1.63	1.76	+2.37
Volume (fractal, $R^{2.4}$)	312	494	545	431	1.49	0.47	-2.95
Volume (sphere, R^3)	159	483	543	381	1.74	0.79	-2.09

Modality caveat — read the next table as inversion fine structure. The Laplace inversion is ill-posed: a smooth continuum and a set of discrete modes can fit the same g_2 within noise, so the number and sharpness of peaks depend on the regularization (λ), the radius grid / size_max, the spline basis, and the sphere form factor — none of which the data uniquely fix. The inversion carries ≈ **18 spline degrees of freedom** — ample latitude to place peaks on a continuum. **Treat the apparent peaks/shoulders below as fine structure of this particular inversion — not established sub-populations — unless they are stable under the size_max / λ sensitivity checks and reproducible across replicates.**

Apparent modes & shoulders (CORENN fine structure — inversion-dependent)

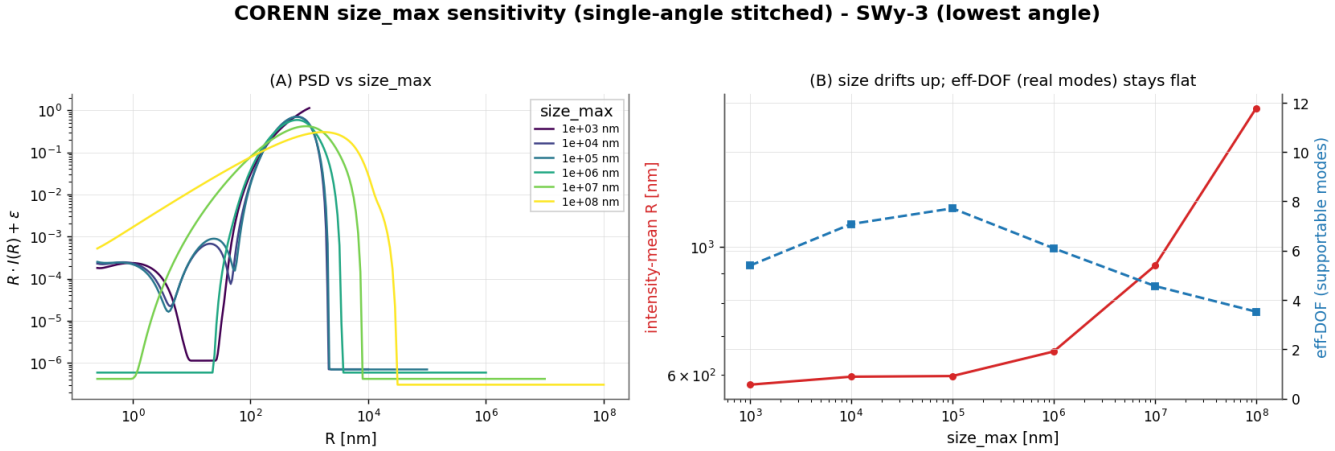
Weighting	Comp.	Peak R (nm)	⟨R⟩ (nm)	Median (nm)	Fraction	GSD	log-skew
Intensity ($\propto R^6$)	1	539	502	511	99%	1.11	-1.93 (tail → small R)
Number (fractal, $R^{4.9}$)	1	57	58	58	55%	1.13	+0.26 (≈ log-symmetric)
	2	157	159	158	24%	1.12	+0.14 (≈ log-symmetric)
	3	513	463	482	21%	1.18	-1.07 (tail → small R)
Number (sphere, R^6)	1	56	57	57	85%	1.13	+0.28 (≈ log-symmetric)
	2	154	157	155	12%	1.12	+0.17 (≈ log-symmetric)
	3	506	449	470	3%	1.20	-0.78 (tail → small R)
Volume (fractal, $R^{2.4}$)	1	527	487	500	91%	1.14	-1.63 (tail → small R)
	2	163	165	163	8%	1.12	+0.05 (≈ log-symmetric)
Volume (sphere, R^3)	1	524	482	497	83%	1.15	-1.52 (tail → small R)
	2	162	163	162	13%	1.12	+0.07 (≈ log-symmetric)
	3	59	60	59	4%	1.13	+0.23 (≈ log-symmetric)

The intensity distribution is led by a peak at **539 nm** (99% of scattered light, tail → small R); under number weighting the dominant component is at **57 nm** — the *same* data, modes relocated and re-weighted by the R-power, which is why no single mean describes it.

Sphere vs fractal correction. The sphere Volume/Number rows assume compact spheres ($V \propto R^3$, $I \propto R^6$); this sample is a **mass fractal** $D_f \approx$ **2.4**, so the fractal rows (volume $R^{2.4}$, number $R^{4.9}$) are the physically-correct conversions — they move every mode's peak/⟨R⟩ upward relative to the sphere rows (the sphere law over-divides the large flocs).

CORENN size_max sensitivity (ill-posedness)

CORENN is ill-posed: where it parks unconstrained mass depends on the **size_max** grid bound. Sweeping size_max on a single-angle stitched inversion shows it directly:



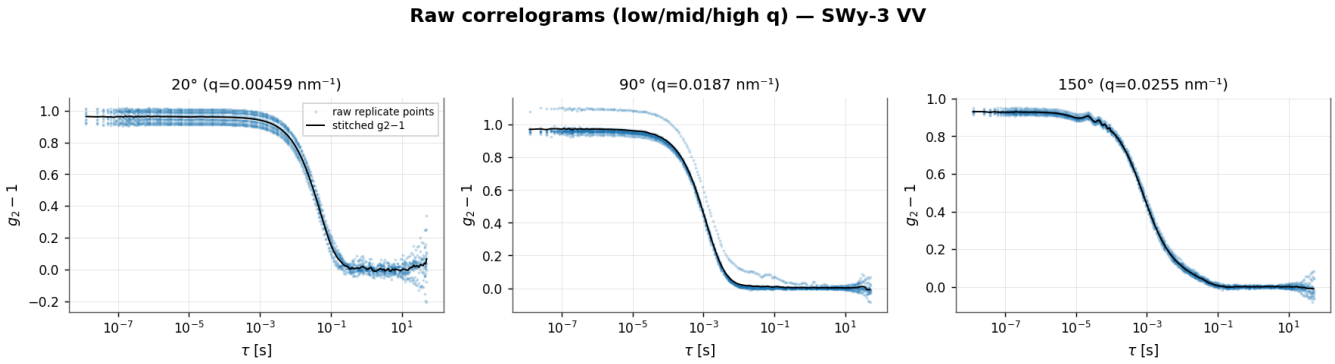
Raising size_max from 1e+03 to 1e+08 nm drifts the intensity-mean size from ˜577 to ˜1739 nm (mass pulled to the unconstrained large-R edge), while the **effective DOF stays ˜ 5.7** – the data support only ˜that many modes either way. So a large/arbitrary size_max (the vendor 3.5e11 nm) buys edge-pile and ghost peaks, not information. The pipeline bounds size_max to the physically resolvable window (pi/q for the multi-angle PSD; the trusted-lag size for the single-angle slow-mode) and reports the effective DOF as the honest cap on resolvable modes.

Data Inventory

One row per analysed condition (per-angle detail is in Per-Curve Fit Results).

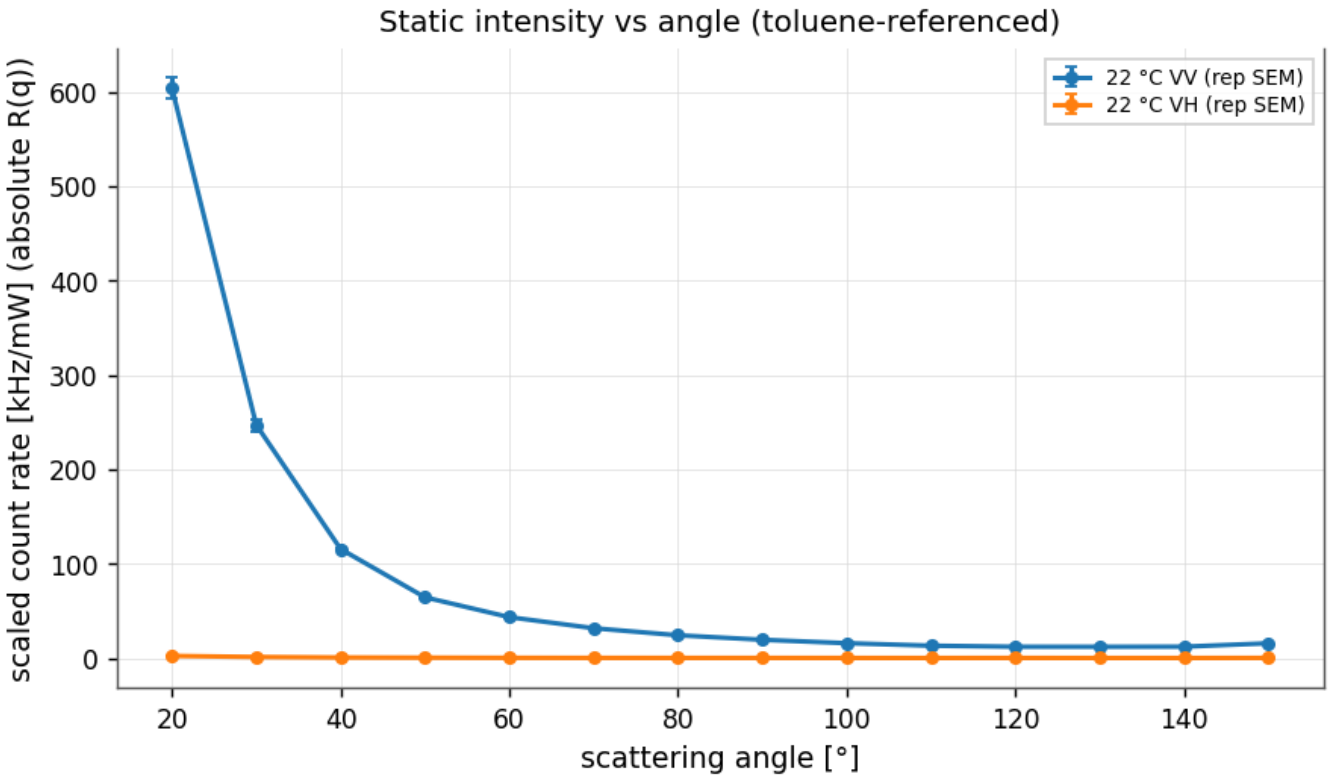
condition	channel	n angles (kept/total)	angle range	q range [nm^{-1}]	T [K]	scaled count rate [kHz/mW]	reps	missing meta
22 °C	VV	14/14	20–150°	0.00459–0.0255	295.1	12–604	14	0

Raw Correlograms



Raw VV correlograms at low/mid/high q: the stitched multi-tau correlation (black) over the semi-transparent raw points.

Count Rates And Static Intensities (absolute, toluene-referenced)



Scaled count rate (absolute Rayleigh ratio) vs angle, VV and VH channels (raw data).

Per-Curve Fit Results

Per-curve fits — 22 °C (VV)

Per-angle 2nd-order cumulant on the leading decay (g1_min=0.6); $R_{\mathrm{h,app}} = \mathrm{kT}\ \mathrm{q}^2/(6\pi\eta\ \Gamma)$.

angle [°]	q [nm^{-1}]	Γ [1/s]	μ_2	PDI	β	$R_{\mathrm{h,app}}$ [nm]	reps	accepted
20	0.00459	10.9	4.8e+01	0.41	0.944	439	10	yes
30	0.00684	31.6	3.1e+02	0.32	0.899	335	10	yes
40	0.00904	68.4	2.0e+03	0.42	0.948	270	10	yes
50	0.0112	122.8	6.2e+03	0.41	0.938	230	10	yes
60	0.0132	187.2	1.4e+04	0.41	0.953	211	10	yes
70	0.0152	262.7	3.1e+04	0.45	0.963	198	10	yes
80	0.017	318.8	6.2e+04	0.61	0.965	205	10	yes
90	0.0187	417.2	8.4e+04	0.48	0.961	189	10	yes
100	0.0202	518.9	1.3e+05	0.46	0.976	179	10	yes
110	0.0216	627.3	1.8e+05	0.46	0.970	169	10	yes
120	0.0229	682.0	2.4e+05	0.53	0.959	174	10	yes
130	0.0239	689.8	3.0e+05	0.63	0.967	188	10	yes
140	0.0248	686.5	3.5e+05	0.73	0.961	203	10	yes
150	0.0255	510.5	2.6e+05	0.99	0.921	289	10	yes

Rejected Data And Failure Diagnostics

No angles or curves were rejected.

Assumption Register

assumption	value	source	affects	sensitivity checked?	comment
wavelength	632.8 nm	instrument default	q, D, R_h	no	—
refractive index	1.33	water, 22 °C	q	no	replace with measured
viscosity	0.95 mPa·s	recorded export (water, 0.95 mPa·s)	R_h	yes	water at 22 °C — physically reasonable recorded value
q source	recalculated from angle	pipeline	all q-dependent fits	no	vendor 'Scattering vector [1/nm]' column is mislabelled (values are 1/m); q recomputed from angle, reported in nm^{-1}
static scale	absolute (toluene-referenced) — assumed, unverified	LSII toluene reference	SLS, R_g	no	toluene reference not confirmable from parsed export
form factor	homogeneous sphere (RGD)	pipeline	PSD radii, ρ check	yes	flag if $\rho \neq 0.78$ or depolarization present
cumulant g1_min	0.6	pipeline config	Γ, PDI	yes	leading-decay window
PSD radius grid	8–560 nm	dynamic observability + static q guardrail	PSD peaks	yes	upper=min(dynamic 560 nm, static 685 nm, default 5000 nm); limiting=dynamic; tau_max=0.026 s, min field decay=0.223
PSD basis	24 B-spline control points	pipeline config	PSD smoothness	yes	—
regularization	GCV-selected λ=1.4e+00	pipeline	PSD width	yes	interior minimum required
baseline	free per fit (CORENN); leading-decay cumulant	pipeline config	Γ, PSD	yes	—

Quality Gates

Provisional report: **11 PASS, 1 WAIVE, 0 FAIL** of 12.

gate	status	note
q physically plausible / reproducible from angle	**PASS**	q recomputed from angle (vendor q column mislabeled 1/nm)
viscosity not a placeholder	**PASS**	recorded value used
fit bounds stated for every fit	**PASS**	cumulant g1_min=0.6; CORENN tau_max=0.026s, Rmax=560 nm (dynamic)
rejected curves/angles listed	**PASS**	0 rejected, tabulated
residual-structure test reported (not r^2 alone)	**PASS**	runs test + lag-1 autocorrelation
headline R_h carries a bootstrap CI	**PASS**	replicate + angle/model bootstrap
hold-out cross-validation reported	**PASS**	leave-one-angle-out
CORENN converged + well-posed regularization	**PASS**	converged=True, GCV interior=True, κ=6e+13
static scale confirmed absolute	**WAIVE**	toluene reference assumed (LSII default), not confirmable from parsed export
PSD basis (intensity/volume/number) stated	**PASS**	intensity-weighted primary, conversions noted
$\rho = R_g/R_h$ and particle model stated	**PASS**	not determined; Porod slope implies mass fractal / branched (D≈2.4) — sphere PSD form factor flagged inconsistent
figure captions state raw vs derived	**PASS**	captions included

Metadata

field	value
Report title	SWy-3 montmorillonite (1 %, 22 °C) — provisional polarized DLS analysis report
Data source	data/SWy3_1
Report generator	reports/swy3_report.py (provisional)
Analyst / owner	automated pipeline
Dataset class	multi-angle, single-temperature, polarized (VH/VV/3D)
Channels present	VV
Temperatures	295.1 K
Angles	20–150° (14 per condition)
q range	0.00459–0.0255 nm^{-1}
Replicates	14 angle-sets with ≥2 replicates
Solvent constants	η per condition (see table); n=1.33, λ=632.8 nm (water, 22 °C)
Static scale	absolute (toluene-referenced) — assumed, unverified
Particle model	mass-fractal clay floc (Porod $q^{-2.4}$ ⇒ mass fractal / branched (D≈2.4)); PSD sphere form factor is an effective approximation
Primary model	CORENN multi-angle PSD (VV) + depolarized VV/VH model
Report limitations	sphere PSD form factor inconsistent with the mass-fractal Porod slope; toluene reference unverified

Reproducibility Appendix

item	value
Command	`python reports/swy3_report.py`
Environment	python 3.13.5, numpy 2.2.5
Git commit (working tree)	91a3b68 (dirty)
Random seeds	bootstraps seed=0 (fixed)
Generated files	SWy3.md, SWy3.pdf, figures/swy3.png, figures/swy3_raw.png, figures/swy3_slowmode.png, figures/swy3_weightings.png, figures/swy3_sizemax.png, figures/swy3_countrate.png, figures/swy3_highangle.png, SWy3.sidecar.json